

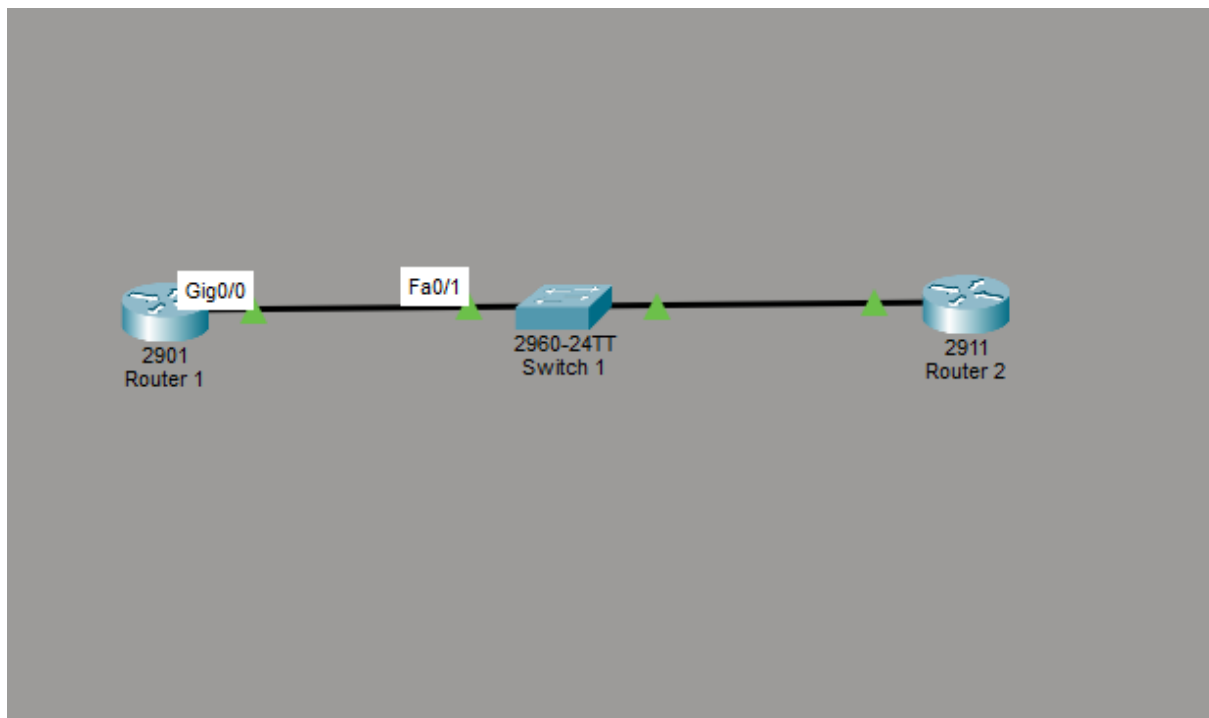


NET512 WEEK 5 PRAC

Lerato Modise



Screenshot of the topology



Hostname R1

```
Router(config-if)#hostname R1
R1(config)#end
R1#
```

Hostname R2

```
up
Router(config-if)#Hostname R2
R2(config)#
```

Hostname SW 1

```
Switch>enable
Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#hostname S1
S1(config)#
```

Configure IP address on R1

```
R1(config)#interface GigabitEthernet0/0
R1(config-if)#ip address 10.10.10.1 255.255.255.0
R1(config-if)#no shutdown
R1(config-if)#
```

Configure IP address on R2

```
Router>enable
Router#
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/!
Router(config)#interface GigabitEthernet0/0
Router(config-if)#ip address 10.10.10.2 255.0.0.0
Router(config-if)#ip address 10.10.10.2 255.0.0.0
Router(config-if)#no shutdown
Router(config-if)#
```

SW 1 Management IP Address

```
S1>enable
S1#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
S1(config)#hostname SW1
SW1(config)#interface vlan 1
SW1(config-if)#ip address 10.10.10.10 255.255.255.0
SW1(config-if)#no shutdown

SW1(config-if)#
%LINK-3-UPDOWN: Interface Vlan1, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan1, changed state to up
```

Ping R2 from SW1

```
SW1#ping 10.10.10.2

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.10.2, timeout is 2 seconds:
.!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 0/9/38 ms
```

CDP Configuration

Neighbour

```
R1#show cdp
Global CDP information:
  Sending CDP packets every 60 seconds
  Sending a holdtime value of 180 seconds
  Sending CDPv2 advertisements is enabled
```

```

R1#show cdp
Global CDP information:
  Sending CDP packets every 60 seconds
  Sending a holdtime value of 180 seconds
  Sending CDPv2 advertisements is enabled
R1#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#interface GigabitEthernet0/0
R1(config-if)#no cdp enable
R1(config-if)#exit
R1(config)#exit
R1#
%SYS-5-CONFIG_I: Configured from console by console

R1#show cdp
Global CDP information:
  Sending CDP packets every 60 seconds
  Sending a holdtime value of 180 seconds
  Sending CDPv2 advertisements is enabled
R1#show cdp neighbor
Capability Codes: R - Router, T - Trans Bridge, B - Source Route Bridge
                  S - Switch, H - Host, I - IGMP, r - Repeater, P - Phone
Device ID         Local Intrfce   Holdtme    Capability   Platform     Port ID
SW1                Gig 0/0         59         S            2960         Fas 0/1

```

Flush CDP

```

Enter configuration commands:
R1(config)#no cdp run
R1(config)#cdp run
R1(config)#end

```

Verify the switch port connected to R2

```

SW1>enable
SW1#show ip interface brief

```

Interface	IP-Address	OK?	Method	Status	Protocol
FastEthernet0/1	unassigned	YES	manual	up	up
FastEthernet0/2	unassigned	YES	manual	up	up

Shutdown the interface connected to R2

```

SW1#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
SW1(config)#interface fa0/2
SW1(config-if)#shutdown

SW1(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/2, changed state to administratively down

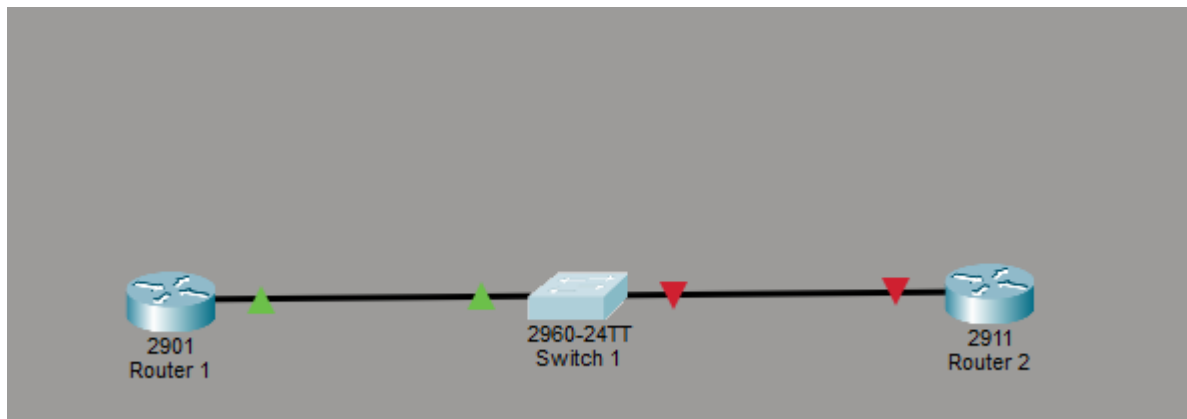
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to down

SW1(config-if)#end
SW1#
%SYS-5-CONFIG_I: Configured from console by console

SW1#show ip interface brief

```

Interface	IP-Address	OK?	Method	Status	Protocol
FastEthernet0/1	unassigned	YES	manual	up	up
FastEthernet0/2	unassigned	YES	manual	administratively down	down



Bring the interface back up

```

SW1#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
SW1(config)#int fa0/2
SW1(config-if)#no shutdown

SW1(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/2, changed state to up

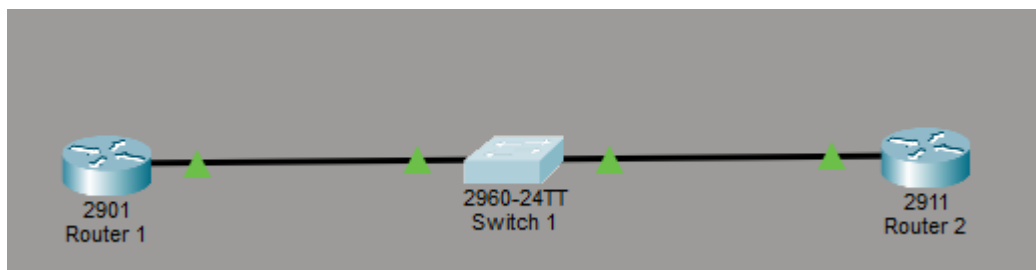
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to up

SW1(config-if)#end
SW1#
%SYS-5-CONFIG_I: Configured from console by console

SW1#show ip interface brief

```

Interface	IP-Address	OK?	Method	Status	Protocol
FastEthernet0/1	unassigned	YES	manual	up	up
FastEthernet0/2	unassigned	YES	manual	up	up



Set the duplex on SW1 interface to half

```
SW1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
SW1(config)#interface fa0/2
SW1(config-if)#duplex half
SW1(config-if)#end
SW1#
%SYS-5-CONFIG_I: Configured from console by console

SW1#show interface fa0/2
FastEthernet0/2 is up, line protocol is up (connected)
  Hardware is Lance, address is 00d0.ffbd.0802 (bia 00d0.ffbd.0802)
  BW 100000 Kbit, DLY 100 usec,
    reliability 255/255, txload 1/255, rxload 1/255
  Encapsulation ARPA, loopback not set
  Keepalive set (10 sec)
  Half-duplex, 100Mb/s
  input flow-control is off, output flow-control is off
  ARP type: ARPA, ARP Timeout 00:04:00
  Last input 00:00:08, output 00:00:05, output hang never
  Last clearing of "show interface" counters never
  Input queue: 0/75/0/0 (size/max/drops/flushes); Total output drops: 0
  Queueing strategy: fifo
  Output queue :0/40 (size/max)
  5 minute input rate 0 bits/sec, 0 packets/sec
  5 minute output rate 0 bits/sec, 0 packets/sec
    956 packets input, 193351 bytes, 0 no buffer
    Received 956 broadcasts, 0 runts, 0 giants, 0 throttles
    0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored, 0 abort
    0 watchdog, 0 multicast, 0 pause input
    0 input packets with dribble condition detected
  2357 packets output, 263570 bytes, 0 underruns
```

Set the duplex to full

```
SW1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
SW1(config)#interface fa0/2
SW1(config-if)#duplex full
SW1(config-if)#end
SW1#
%SYS-5-CONFIG_I: Configured from console by console

SW1#show interfaces fa0/2
FastEthernet0/2 is up, line protocol is up (connected)
  Hardware is Lance, address is 00d0.ffbd.0802 (bia 00d0.ffbd.0802)
  BW 100000 Kbit, DLY 100 usec,
    reliability 255/255, txload 1/255, rxload 1/255
  Encapsulation ARPA, loopback not set
  Keepalive set (10 sec)
  Full-duplex, 100Mb/s
  input flow-control is off, output flow-control is off
  ARP type: ARPA, ARP Timeout 00:04:00
  Last input 00:00:08, output 00:00:05, output hang never
  Last clearing of "show interface" counters never
  Input queue: 0/75/0/0 (size/max/drops/flushes); Total output drops: 0
  Queueing strategy: fifo
  Output queue :0/40 (size/max)
  5 minute input rate 0 bits/sec, 0 packets/sec
  5 minute output rate 0 bits/sec, 0 packets/sec
    956 packets input, 193351 bytes, 0 no buffer
    Received 956 broadcasts, 0 runts, 0 giants, 0 throttles
    0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored, 0 abort
    0 watchdog, 0 multicast, 0 pause input
    0 input packets with dribble condition detected
  2357 packets output, 263570 bytes, 0 underruns
```

Set the speed to 10 mbps

```
SW1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
SW1(config)#interface fa0/2
SW1(config-if)#speed 10
SW1(config-if)#end
SW1#
%SYS-5-CONFIG_I: Configured from console by console

SW1#show ip interface brief
```

Interface	IP-Address	OK?	Method	Status	Protocol
FastEthernet0/1	unassigned	YES	manual	up	up
FastEthernet0/2	unassigned	YES	manual	up	up

Check whether the interface on R2 is still operational

```
R2>enable
R2#show ip interface brief
```

Interface	IP-Address	OK?	Method	Status	Protocol
GigabitEthernet0/0	10.10.10.2	YES	manual	up	up
GigabitEthernet0/1	unassigned	YES	unset	administratively down	down
GigabitEthernet0/2	unassigned	YES	unset	administratively down	down
Vlan1	unassigned	YES	unset	administratively down	down

47. A duplex mismatch causes collisions packet loss.